

A Containerised Development Environment *for* Hyperledger Fabric

This article is a complete primer on how to set up the Hyperledger Fabric development environment on a local machine using Vagrant on an Ubuntu image. It also covers how to deploy Fabric by using Docker Compose to launch Docker containers.



The Hyperledger Fabric development environment uses either Vagrant running an Ubuntu image (which in turn launches Docker containers), or simply deploys the Fabric network with Docker Compose (which also launches Docker containers). This model allows developers to leverage their favourite OS/editors and execute the system in a controlled environment that is consistent across the development team.

Before we begin, if you haven't already done so, you may wish to check that you have all the prerequisites listed below installed on the platform(s) on which you'll be developing blockchain applications and operating Hyperledger Fabric.

- Download the latest version of the *cURL* tool if it is not already installed or if you get errors running the *curl* commands from the documentation.
- Install Docker and Docker Compose.
- Install Vagrant if you wish to set up a development environment with it.
- Hyperledger Fabric uses the Go programming language for many of its components. For development, Go version 1.11.x is required.
- If you wish to develop applications for Hyperledger Fabric leveraging its SDK, you need supporting packages for SDK installed; e.g., for Node.js, you will need to have version 8.9.x of Node.js installed.

- If you are developing on Windows 7, you will want to work within the Docker Quickstart terminal, which uses Git Bash and provides a better alternative to the built-in Windows shell.
- On Windows 10, you should use the native Docker distribution and you may use the Windows PowerShell.

Setting up Git

Before running any Git clone commands, run the following:

```
git config --global core.autocrlf false
git config --global core.longpaths true
```

You can check the setting of these parameters with the following commands:

```
git config --get core.autocrlf
git config --get core.longpaths
```

These need to be false and true, respectively.

Set your GOPATH

Make sure you have properly set up your host's GOPATH environment variable. This allows for building within the host, the VM and containers using the Fabric tools container.